A Locally Optimized Model for Hyperspectral and Multispectral Images Fusion

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***Abstract*— The maintenance of spectral variability between subclass objects and the relationship between hyperspectral (HS) bands have been a fundamental but challenging problem for fusing low spatial resolution (LR) HS and high spatial resolution (HR) multispectral (MS) images. This article presents a locally optimized image segmentation fusion (LOISF) framework for HS super-resolution reconstruction. First, LR HS and HR MS are clustered and segmented, and the label attributes of the segmented objects are identified by the prior information. Then, a novel joint fusion model for different typical ground objects is constructed based on spectral unmixing. The fusion problem is formulated mathematically as a convex optimization of a Frobenius norm, which includes spatial, spectral, and index constraints, with an alternating-directions' optimization featuring linearization providing the solution. Experimental results demonstrate that the proposed LOISF preserves both spatial details and texture, achieving high spectral fidelity, and yielding significantly improved image quality compared to other state-of-the-art fusion methods.**

***Index Terms*— Hyperspectral (HS), image fusion, locally optimized image segmentation fusion (LOISF), multispectral, unmixing.**

I. Introduction

HYPERSPECTRAL (HS) data have a fine spectral resolution, which makes each band have limited absorption of the reflected energy on the ground surface, and obtain low spatial resolution (LR) [1]. The spectral resolution of MS data is low, but each band can fully reflect the distribution of reflected energy on the ground surface, and MS data achieve a high spatial resolution (HR) [2]. However, numerous work needs to use HR HS data, and many studies are committed to multisource data fusion in recent years to achieve information complementation of multisource data [3], [4].

The existing fusion methods have achieved comparative fusion performance. However, the reconstructed image always has limited spectral fidelity in subclass ground objects. For example, different vegetations affected by "different objects with the same spectrum" have various spectral distortions, which seriously affects the accuracy of ground object identification in realistic applications. Besides, the band relations are weakly constrained during the process of image reconstruction, which may cause serious spectral distortion in the fused images [5].

Aiming at the above two problems, in this article, we propose a locally optimized image segmentation fusion (LOISF) framework for HS and MS fusion. This idea of our framework is to cluster the MS images, and when there are a lot of small patches, the mathematical morphology operator, dilation and erosion, is used for reclustering, and the HS and MS images are segmented according to the clustering map. Second, the mean spectral curves of the segmented map are used to match the USGS spectral library, and the fusion models are determined. Finally, different fusion models are executed for different objects, and the final fused image is obtained through the mosaic of each fused image. For these fusion models, we add different index constraints based on spectral and spatial constraints for different objects (the index needs band operation, and the relationship between LR HS bands can be well transferred to the fused image by index constraint). For example, increase the normalized difference vegetation index (NDVI) constraints for vegetation areas [6], and add the normalized difference water index (NDWI) constraints for water areas [7]. The main contributions of LOISF are given as follows.

1) LOISF first joint considers the influence of spectral variability of subclass ground objects and band relationship retention on the fusion image quality. It could effectively improve the spatial resolution of LR HS and reduce the fusion error of subclass ground objects and spectral distortion of the final fused image.
2) LOISF establishes the high-fidelity fusion model by comprehensively considering spectrum, space, and index

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constraint, which enables different ground objects to simultaneously obtain high fusion quality and effectively reduces the spectral and spatial distortions of the fused image.

3) LOISF has fewer parameters and robust performance. The LOISF can not only achieve comparative performance in images with simple ground objects but also obtain the best results for these images with more complicated underlying surfaces compared with the current advanced methods, which shows that our method is robust.

The remainder of this article is organized as follows. Section II introduces the background work. Section III gives a specific description of LOISF. Section IV analyzes the experimental results. The conclusion is finally provided in Section V.

II. Related Works

HS and MS fusion methods are mainly divided into three categories: pansharpening-based methods, machine-learning-based methods, and deep-learning-based methods.

A. Pansharpening-Based Methods

The studies of data fusion are initially focused on pansharpening methods. These methods include component substitution (CS)-based methods and multiresolution analysis (MRA)-based methods [8], [9], and these methods are used to fuse MS and panchromatic (PAN) data. Vivone *et al.* [8] and Sun *et al.* [11], respectively, combined band grouping and regression synthesis method with these pansharpening methods to fuse HS and MS data [8], [10]. CS-based methods assume that spatial and spectral information of the HS data can be separated, and the spatial resolution can be enhanced by replacing the spatial information of HS data with MS images [11]. For example, Carper *et al.* [12] converted LR image from red–green–blue (RGB) space to intensity–hue–saturation (IHS) space and replaced the intensity component of LR image with HR image to achieve fusion. Kwarteng and Chavez [13] used HR image to replace the first principal component of the LR image, and the method achieved information complementation. Shettigara [14] summarized these methods and proposed a general CS framework and showed that the main differences of current methods were the extraction method of spatial information and the design of the gain matrix. MRA-based methods assume that the main difference between HR image and LR image is the spatial information, and the missing spatial information of the LR image can be extracted from the HR image and added to achieve image fusion [15]. Chavez *et al.* [16] obtained the missing spatial information of LR data from HR data by using a high-pass filter and used the identity matrix to inject the spatial information into the LR image. Aiazzi *et al.* [17] introduced the pyramid decomposition theory and proposed modulation transfer function-generalized Laplacian pyramid (MTF-GLP); Vivone *et al.* [18] introduced wavelet decomposition theory into the field of image processing and proposed "À trous" wavelet transform (ATWT) for image fusion. Vivone *et al.* [8] proposed a general MRA framework by summarizing these methods and believed that the difference between these methods was the multiscale decomposition method and the design of the gain matrix. Pansharpening-based methods are widely used and have high computational efficiency [19]. However, the fusion performance is limited [20]. For example, CS-based methods have severe spectral distortion [21], and MRA-based methods have limited spatial resolution enhancement [22].

B. Machine-Learning-Based Methods

In recent years, the machine learning theory has been widely applied in remote sensing image processing, and sparse representation and image unmixing theory have been used as typical matrix factorization methods for HS and MS image fusion [5], [23], [24]. These methods assume that the image can be decomposed into spectral and spatial matrices, and HR HS can be obtained by reorganizing the spectral matrix of the HS image and the spatial matrix of the MS image [25], [26]. For example, Kawakami *et al.* [27] combined spectral dictionary learning and sparse coding algorithms to achieve spatial resolution enhancement. Yokoya *et al.* [28] used coupled nonnegative matrix factorization (CNMF) to extract the endmember and abundance matrix of HS and MS data, and reorganized the HS endmember matrix and MS abundance matrix to realize image fusion. Nezhad *et al.* [23] combined sparse representation and image unmixing theory to realize the information reorganization between images. In addition, tensor decomposition as high-order matrix decomposition method is also widely used in data fusion [29]. These methods achieve image fusion by constructing a tensor kernel to reorganize two spatial matrices of MS data and a 1-D matrix of HS data [30]. Typical methods include the nonlocal sparse tensor factorization (NLSTF) proposed by Dian *et al.* [31] using the tucker decomposition method and the fusion method of canonical polyadic (CP) decomposition proposed by Xu *et al.* [32]. Machine-learning-based methods obtain comparative fusion performance, but these methods usually have more parameters, lower robustness, and computational efficiency [33].

C. Deep-Learning-Based Methods

At present, deep learning methods have emerged and are rapidly used in image processing due to the unique advantages in solving nonlinear problems [34], [35], and they are regarded as the new generation. Deep-learning-based methods usually need to pretrain the relationship between the input data and the target data, and map the relationship to another input data to predict the output data [36], [37]. For example, Yang *et al.* [38] proposed two-branch convolutional neural networks (Two-CNN) to extract the spectral information of HS image and the spatial information of MS image and implemented feature fusion through a fully connected layer to achieve image fusion. Palsson *et al.* [39] found that the 3-D data converting to 2-D data may lose spatial information and proposed 3-D CNN for image fusion. Zhou *et al.* [40] assumed that there is different feature information in different scales of image and combined Laplace decomposition and CNN to propose a pyramid fully convolutional network (PFCN)

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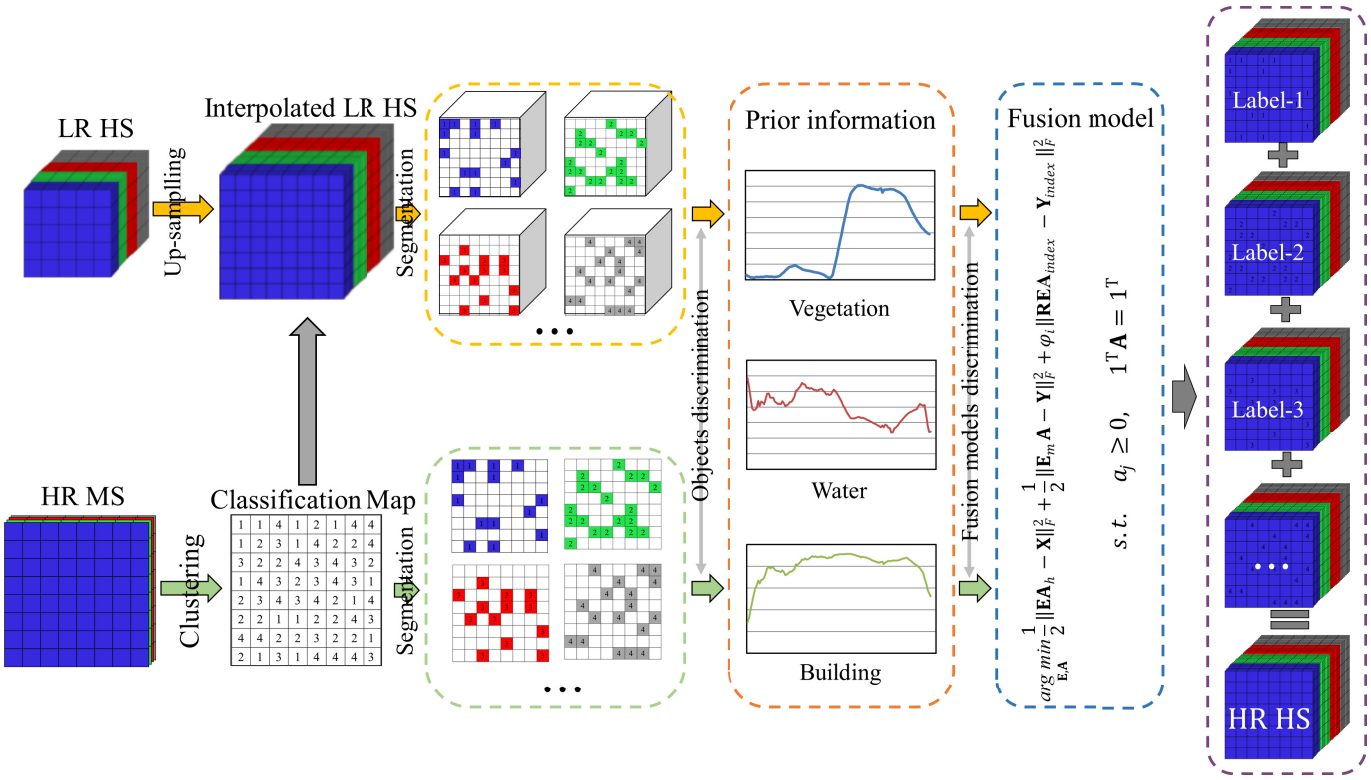

Fig. 1. Flowchart of the LOISF method.

for data fusion [40]. Deep learning methods achieve better fusion results. However, these methods require a large amount of training data and have low computational efficiency and overfitting problems [41], [42].

In realistic applications, these pansharpening-based methods have limited fusion performance, but deep-learning-based methods have low computational efficiency [43], [44]. Machine-learning-based methods have comparative advantages due to the better fusion result, calculation efficiency, and algorithm mechanism, especially unmixing-based methods [1].

III. LOISF

A. Idea of LOISF

For HS and MS image fusion, the maintenance of spectral variability between subclass objects and the relationship between HS bands is important and challenging. Based on the above motivation, we proposed the LOISF for HS and MS fusion, as shown in Fig. 1. $\mathbf{X} \in \mathbb{R}^{n\times m\times B}$ is the LR HS, where n and m are the height and width, respectively, and B is the number of bands. $\mathbf{Y} \in \mathbb{R}^{N\times M\times b}$ represents HR MS data and $n < N, m < M$, and $B > b$. $\tilde{\mathbf{X}}$ and $\mathbf{Z}$ represent the unsampled LR HS and fused images, respectively.

The LOISF consists of three parts: image clustering and discrimination, the LOISF fusion model, and optimization solution. For the image clustering and discrimination, we first use MS data for unsupervised clustering and use mathematical morphology operators, dilation and erosion, for reclustering when there are a lot of small patches. Then, the upsampling HS image is segmented by the clustered map of MS to obtain multiple datasets of homogeneous regions. For the discrimination of fusion models, we use the USGS spectral library as *a priori* information to calculate the mean spectral curves of vegetation, water, and building and calculate the mean spectral curves of all pixels in the segmented HS. By using spectral angle measurement (SAM) [15], we calculate the correlation between the mean spectral curves and determine the category of the objects. For the LOISF fusion model, local index constraints are introduced to establish the high-fidelity fusion model: 1) establish space-spectrum-NDVI constraint for vegetation area fusion; 2) establish space-spectrum-NDWI constraint for water area fusion; 3) establish space-spectrum-simulated normalized difference building index (SNDBI) constraint for building area fusion [45]; and 4) only establish generalized space-spectrum constraint for other objects. According to different objects, LOISF selects the corresponding fusion models to obtain these fused segmentation images. After the above procedure, we mosaiced these fused images together to achieve the final HR HS.

B. Image Clustering and Discrimination

The proposed LOISF uses a hierarchical sparse subspace-based clustering algorithm (HESSC) [46] for image clustering. HESSC selects a random sample for binary clustering based on the lasso-based binary clustering algorithm, and the entropy-based ensemble algorithm is used to reduce the effect of random selection. In addition, in order to have an automatic way to generate the number of clusters, the standard is defined by comparing the calculated reconstruction error value at the parent node and the child node. Finally, assign labels to each terminal node to generate the final cluster map.

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Besides, we use the USGS spectral library as *a priori* information and calculate the mean spectral curves of vegetation, water bodies, and building. We also calculate the mean spectral curves of all pixels in the segmented image. The SAM is used to calculate the spectral difference between the prior mean spectral curves and the mean spectral curves of the segmented map, and the final fusion model is determined by setting the threshold. The specific algorithm is given as follows:

$$\mathrm{V}_{\mathrm{mcurve}} = \mathrm{mean}(\mathrm{USGS}_{\mathrm{Vegetation}}) \tag{1}$$

$$W_{\mathrm{mcurve}} = \mathrm{mean}(\mathrm{USGS}_{\mathrm{Water}}) \tag{2}$$

$$\mathrm{B}_{\mathrm{mcurve}} = \mathrm{mean}(\mathrm{USGS}_{\mathrm{Building}}) \tag{3}$$

where $\mathrm{USGS}_{\mathrm{Vegetation}}$, $\mathrm{USGS}_{\mathrm{Water}}$, and $\mathrm{USGS}_{\mathrm{Building}}$ denote the spectral curves of vegetation, water, and building in the USGS spectral library, respectively. $\mathrm{V}_{\mathrm{mcurve}}$, W_{mcurve}, and $\mathrm{B}_{\mathrm{mcurve}}$ represent the corresponding mean spectral curves of different objects, respectively. mean$(\cdot)$ is the mean operation

$$\mathrm{Label}_{\mathrm{mcurve}} = (P_1 + P_2 + \cdots + P_k)/k \tag{4}$$

in which $\mathrm{Label}_{\mathrm{mcurve}}$ represents the mean spectral curve of all pixels in the segmented image, P is the pixel of the segmented image, and k is the pixel number. The discriminant formula is set as follows:

$$\phi_1 = \frac{\mathrm{V}_{\mathrm{mcurve}}^T \mathrm{Label}_{\mathrm{mcurve}}}{\|\mathrm{V}_{\mathrm{mcurve}}\|_2 \|\mathrm{Label}_{\mathrm{mcurve}}\|_2} \tag{5}$$

$$\phi_2 = \frac{W_{\mathrm{mcurve}}^T \mathrm{Label}_{\mathrm{mcurve}}}{\|W_{\mathrm{mcurve}}\|_2 \|\mathrm{Label}_{\mathrm{mcurve}}\|_2} \tag{6}$$

$$\phi_3 = \frac{\mathrm{B}_{\mathrm{mcurve}}^T \mathrm{Label}_{\mathrm{mcurve}}}{\|\mathrm{B}_{\mathrm{mcurve}}\|_2 \|\mathrm{Label}_{\mathrm{mcurve}}\|_2} \tag{7}$$

where ϕ_1, ϕ_2, and ϕ_3 are the SAM values of the prior mean spectral curves of vegetation, water, and building with the mean spectral curves of the segmented maps, respectively,

$$\begin{cases} \text{vegetation}, & \textbf{if } \phi_1 < \phi_2 || \phi_1 < \phi_3 || \phi_1 < \epsilon \\ \text{water}, & \textbf{if } \phi_2 < \phi_1 || \phi_2 < \phi_3 || \phi_2 < \epsilon \\ \text{building}, & \textbf{if } \phi_3 < \phi_1 || \phi_3 < \phi_2 || \phi_3 < \epsilon \\ \text{other}, & \textbf{else} \end{cases} \tag{8}$$

in which ϵ is the threshold and $\epsilon = 1^{-7}$.

C. LOISF Fusion Model

Based on the spectral and spatial relationships between the LR HS, HR MS observations, and the ideal HR HS fused images, the LR HS can be seen as a spatial downsampling and fuzzy version of the HR HS, and the HR MS represents the spectral downsampling version of HR HS [5]. Therefore,

$$\mathbf{X} = \mathbf{SZ} \tag{9}$$

$$\mathbf{Y} = \mathbf{RZ} \tag{10}$$

where $\mathbf{S}$ represents spatial response function, $\mathbf{R}$ denotes the spectral response function, and the spectral response function of MS data is generally used.

According to the theory of HS mixed pixel decomposition, a pixel can be obtained by multiplying multiple pure endmembers and an abundance matrix [43]. Therefore,

$$\mathbf{Z} = \mathbf{EA} \tag{11}$$

in which $\mathbf{E}$ is the endmember matrix of $\mathbf{Z}$ and $\mathbf{E} = [e_1, e_2, \ldots, e_D]$, $\mathbf{A}$ is the abundance matrix of $\mathbf{Z}$, and D is the number of endmembers.

By introducing formula (11) into formulas (9)–(10), we can get

$$\mathbf{X} = \mathbf{EA}_h, \quad \mathbf{A}_h = \mathbf{SA} \tag{12}$$

$$\mathbf{Y} = \mathbf{E}_m\mathbf{A}, \quad \mathbf{E}_m = \mathbf{RE} \tag{13}$$

where $\mathbf{A}_h$ is the abundance matrix of $\mathbf{X}$ and $\mathbf{E}_m$ is the endmember matrix of $\mathbf{Y}$.

Through the above relationship, the general fusion formulation can be constructed

$$\begin{aligned} &\arg\min_{\mathbf{E},\mathbf{A}} \ \frac{1}{2}\|\mathbf{EA}_h - \mathbf{X}\|_F^2 + \frac{1}{2}\|\mathbf{E}_m\mathbf{A} - \mathbf{Y}\|_F^2 \\ &\text{s.t. } a_{ij} \geq 0, \quad 1^T\mathbf{A} = 1^T \end{aligned} \tag{14}$$

where $\|\cdot\|_F^2$ is the Frobenius norm, ij is the coordinate of the elements of $\mathbf{A}$, $a_{ij} \geq 0$ is the nonnegative constraint of abundance matrix, and $1^T\mathbf{A} = 1^T$ represents a constraint where the sum of the abundance matrix is one.

However, the above fusion model globally extracts spectral information, and this extracted spectral information is limited and unrepresentative when the objects are complicated. For example, only one endmember is extracted for a variety of vegetation. As a result, the fused image cannot reflect the spectral variability among subclass objects, which makes the fused image have spectral distortion and seriously affects the application performance, such as fine objects recognition. Therefore, we cluster the images and classify all the segmented images into four categories, vegetation, water, building, and others, and use local constraints to reduce the error caused by global constraints. At the same time, for different objects, index constraints are added based on the general formulation (14) to improve the fusion performance from the perspective of band relation transfer. The specific algorithm is given as follows:

$$\begin{aligned} &\arg\min_{\mathbf{E},\mathbf{A}} \ \frac{1}{2}\|\mathbf{EA}_h - \mathbf{X}\|_F^2 + \frac{1}{2}\|\mathbf{E}_m\mathbf{A} - \mathbf{Y}\|_F^2 \\ &\qquad + \varphi_i\|\mathbf{REA}_{\mathrm{index}} - \mathbf{Y}_{\mathrm{index}}\|_F^2 \\ &\text{s.t. } a_j \geq 0, \quad 1^T\mathbf{A} = 1^T \end{aligned} \tag{15}$$

where $\mathbf{REA}_{\mathrm{index}}$ represents the indexes of $\mathbf{REA}$, $\mathbf{Y}_{\mathrm{index}}$ is the indexes of $\mathbf{Y}$, index $=$ [NDVI, NDWI, SNDBI], φ_i is the regularization coefficient of the ith ground object, $i =$ [vegetation (NDVI), water (NDWI), building (SNDBI), Other objects], and $\varphi_i = 0$ when i represents other objects.

D. Optimization Solution

In CNMF, Yokoya *et al.* [28] introduced multiplicative update rules that are guaranteed to converge to local optima under the nonnegativity constraints of two factorized matrices. In this article, the multiplicative update rules are introduced for the update of $\mathbf{E}$ and $\mathbf{A}$. We use the simplex identification via split augmented Lagrangian (SISAL) [47] to update endmember, and SISAL is one of the convex geometry theory-based endmember extraction methods and has obtained comparative

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Algorithm LOISF Method for Hyperspectral Super-Resolution Reconstruction

Input: LR HS $\mathbf{X}$, HR MS $\mathbf{Y}$ and define the maximum number of iterations as *ite,* the number of clusters as F, ϕ_1, ϕ_2 and ϕ_3 are the SAM values of the prior mean spectral curves of objects with the mean spectral curves of the segmented maps, respectively.
Output: HR HS $\mathbf{Z}$

Step 1. Data preprocessing—HESSC images segmentation
Step 2. Discriminate model

$$\begin{cases} vegetation, & \textbf{if } \phi_1 < \phi_2 || \phi_1 < \phi_3 || \phi_1 < \epsilon \\ water, & \textbf{if } \phi_2 < \phi_1 || \phi_2 < \phi_3 || \phi_2 < \epsilon \\ building, & \textbf{if } \phi_3 < \phi_1 || \phi_3 < \phi_2 || \phi_3 < \epsilon \\ other, & \textbf{else} \end{cases}$$

Step 3. Iterative optimization
for i = 1:F
1) Solving $\arg\min_{\mathbf{E},\mathbf{A}_h} \frac{1}{2}\|\mathbf{E}\mathbf{A}_h - \mathbf{X}\|_F^2$:
Initialize $\mathbf{E}$ by VCA and update $\mathbf{A}_h$ by (17), with $\mathbf{E}$ fixed.
Update $\mathbf{E}$ by (16), with $\mathbf{A}_h$ fixed.
if $\mathbf{E}\mathbf{A}_h - \mathbf{X} < \varepsilon_1 || \ ite > 200$ **end**
Output: $\mathbf{E}$, $\mathbf{A}_h$
2) Solving $\arg\min_{\mathbf{E}_m,\mathbf{A}} \frac{1}{2}\|\mathbf{E}_m\mathbf{A} - \mathbf{Y}\|_F^2$:
Initialize $\mathbf{E}_m$ by $\mathbf{E}_m = \mathbf{R}\mathbf{E}$ and update $\mathbf{A}$ by (21), with $\mathbf{E}_m$ fixed.
Update $\mathbf{E}_m$ by (20), with $\mathbf{A}$ fixed.
if $\mathbf{E}_m\mathbf{A} - \mathbf{Y} < \varepsilon_2 || \ ite > 200$ **end**
Output: $\mathbf{E}_m$, $\mathbf{A}$
3) Solving $\arg\min_{\mathbf{E},\mathbf{A}} \varphi\|\mathbf{R}\mathbf{E}\mathbf{A}_{Index} - \mathbf{Y}_{Index}\|_F^2$:
if $\mathbf{R}\mathbf{E}\mathbf{A}_{Index} - \mathbf{Y}_{Index} < \varepsilon_3 || \ ite > 200$ **end**
else $\mathbf{A}_h = \mathbf{A}\mathbf{S}$ and update $\mathbf{E}$ by (16)
end
Output: $\mathbf{E}$, $\mathbf{A}$
end
$\mathbf{Z} = \mathbf{E} * \mathbf{A}$

performance in HS unmixing, and we use sparse unmixing by variable splitting and augmented Lagrangian (SUNSAL) [48] for abundance estimation and update. Equations (16) and (20) show the specific usage of SISAL, and SUNSAL is used in (17) and (21), which are given as

$$\mathbf{E} \leftarrow \arg\max_{\mathbf{E}} \ \log|\det(\mathbf{E})| + \lambda 1^T \text{hinge}\left(\mathbf{A}_h + \mathbf{E}^{-1} w\right) 1_n \tag{16}$$

$$\mathbf{A}_h \leftarrow \arg\min_{\mathbf{A}_h} \ \frac{1}{2}\|\mathbf{X} - \mathbf{E}\mathbf{A}_h\|_2^2 + \lambda\|\mathbf{A}_h\|_1 + \mathcal{L}_{\mathbb{R}+}(\mathbf{A}_h) \tag{17}$$

$$\mathbf{E}\mathbf{A}_h - \mathbf{X} < \varepsilon_1 \tag{18}$$

in which T represents transpose operation. n is the number of endmembers, hinge$(\cdot)$ represents a piecewise function, $\mathcal{L}_{\mathbb{R}+}$ is the indicator function, λ is the regularization coefficient, w is the projection vector [48], and $\varepsilon_1 = 1e-8$ is the threshold.

In the initialization stage, the virtual dimension (VD) method [49] is used to evaluate the number of endmembers of LR HS, and the vertex component analysis (VCA) [47] is used to obtain the initial endmember matrix $\mathbf{E}$ of LR HS. Then, the initial endmember matrix $\mathbf{E}$ is brought into (17) to obtain $\mathbf{A}_h$, $\mathbf{A}_h$ is brought into (16) to update $\mathbf{E}$, the updated $\mathbf{E}$ is also brought into (17) to update $\mathbf{A}_h$, and loop until the obtained $\mathbf{E}$ and $\mathbf{A}_h$ meet the convergence conditions of formula (18) to stop the iteration

$$\mathbf{E}_m = \mathbf{R}\mathbf{E} \tag{19}$$

$$\mathbf{E}_m \leftarrow \arg\max_{\mathbf{E}_m} \ \log|\det(\mathbf{E}_m)| + \lambda 1^T \text{hinge}\left(\mathbf{A} + \mathbf{E}_m^{-1} w\right) 1_n \tag{20}$$

$$\mathbf{A} \leftarrow \arg\min_{\mathbf{A}} \ \frac{1}{2}\|\mathbf{Y} - \mathbf{E}_m\mathbf{A}\|_2^2 + \lambda\|\mathbf{A}\|_1 + \mathcal{L}_{\mathbb{R}+}(\mathbf{A}) \tag{21}$$

$$\mathbf{E}_m\mathbf{A} - \mathbf{Y} < \varepsilon_2. \tag{22}$$

When (16)–(18) are satisfied, the optimized $\mathbf{E}$ is brought into (19) to obtain the initialized $\mathbf{E}_m$, $\mathbf{E}_m$ is brought into (21) to obtain $\mathbf{A}$, $\mathbf{A}$ is brought into (20) to update $\mathbf{E}_m$, and the updated $\mathbf{E}_m$ is brought into (21) to update $\mathbf{A}$, and loop until the obtained $\mathbf{E}_m$ and $\mathbf{A}$ meet the convergence conditions $\varepsilon_2 = 1e-8$ of formula (22) to stop the iteration

$$\text{CC}(\mathbf{Y}_{\text{index}}, \mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}) = \frac{\sum_{j=1}^{n}\left(\mathbf{Y}_{\text{index}}^j - \mu_{\mathbf{Y}_{\text{index}}}\right)\left(\mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}^j - \mu_{\mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}}\right)}{\sqrt{\sum_{j=1}^{n}\left(\mathbf{Y}_{\text{index}}^j - \mu_{\mathbf{Y}_{\text{index}}}\right)^2 \sum_{j=1}^{n}\left(\mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}^j - \mu_{\mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}}\right)^2}} < \varepsilon_3 \tag{23}$$

$$\mathbf{A}_h = \mathbf{S}\mathbf{A} \tag{24}$$

in which CC$(\cdot)$ is the correlation coefficient between these images [1]. j represents the jth pixel, and n is the number of pixels. $\mu_{\mathbf{Y}_{\text{index}}}$ and $\mu_{\mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}}$ represent the sample mean of $\mathbf{Y}_{\text{index}}$ and $\mathbf{R}\mathbf{E}\mathbf{A}_{\text{index}}$, respectively, $\varepsilon = 1e-4$ is the threshold.

After the above iteration is completed, the optimized $\mathbf{E}$ and $\mathbf{A}$ are used to reconstruct the image, and the reconstructed image is spectrally downsampled. CC$(\cdot)$ is used to calculate the index correlation between images. When (23) is not satisfied, (24) is used to obtain $\mathbf{A}_h$, and $\mathbf{A}_h$ is brought into (16) for recirculation until the convergence condition ε_3 is met.

IV. Experiments

A. Experimental Data

LOISF is verified through four datasets, including two simulated datasets and two real datasets. Table I shows the specific information of these datasets, and Fig. 2 shows the prior spectral curves from USGS spectral library.

1) Chikusei Datasets: Chikusei HS data are produced and published by The University of Tokyo (https://naotoyokoya.com/Download.html), captured by the Headwall Hyperspec-VNIR-C sensor in Chikusei, Japan, on July 29, 2014. The spectral range is 343–1018 nm, the spatial resolution is 2.5 m, and a total of 122 bands are used in the experiment as the reference image. The spectral response function of GF-1 is selected to produce HR MS, and the Gaussian filter and the downsampling factor of 4 are used to obtain LR HS. This image scene includes different types of crops, a variety of vegetation, buildings of different materials, and artificial and natural water bodies and represents a typical farming area with complex ground objects.

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TABLE I
CHARACTERISTIC OF SIMULATED DATASETS AND REAL DATASETS

Categories	Dataset	Instrument	Spatial resolution (m)	Semi Synthetic	Dimensions	Bands
Simulated Datasets	Chikusei	—	2.5	HS	250*250	120
				MS (GF-1)	1000*1000	4
	DFC2018 Houston	—	1	HS	400*400	48
				MS (SPOT-4)	1200*1200	4
Realistic Datasets	Yellow River Estuary	GF-5	30	HS	240*240	180
		Sentinel-2A	10	MS	720*720	4
	Poyang Lake	GF-5	30	HS	210*210	160
		GF-1	7.5	MS	840*840	4

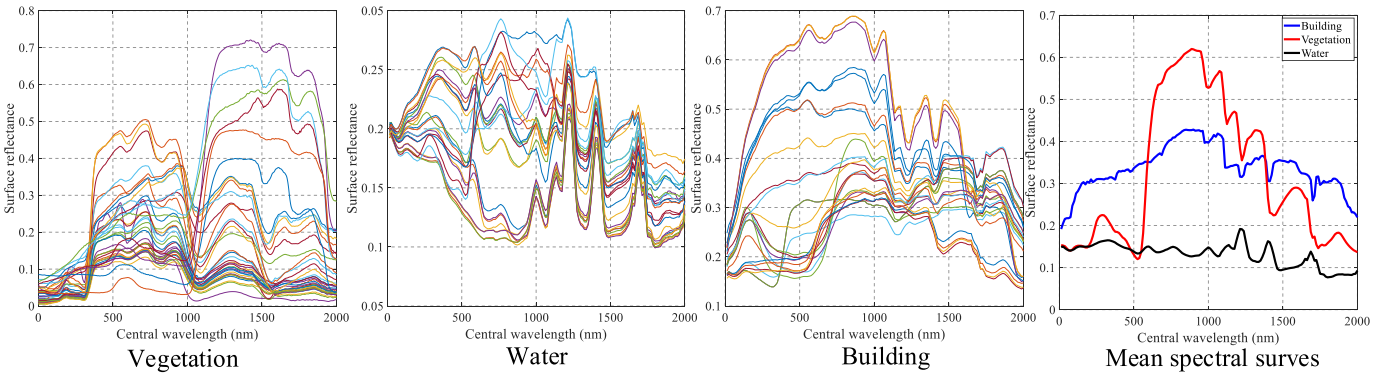

Fig. 2. Prior spectral curve from the USGS spectral library.

2) DFC2018 Houston Dataset: DFC2018 Houston HS data are produced by Dr. Saurabh Prasad from the Laboratory of the University of Houston (http://hyperspectral.ee.uh.edu/page). It contains 48 bands, and the spatial resolution is 1 m. The spectral response function of SPOT-4 is selected to produce HR MS, and the Gaussian filter and the downsampling factor of 3 are used to obtain LR HS. The DFC2018 Houston dataset is typical urban areas with complex underlying surface conditions, and the main ground objects include different types of vegetation, buildings of different materials, and urban water bodies.

3) Yellow River Estuary Dataset: Yellow River Estuary HS data are from Gaofen (GF)-5 HS sensor and downloaded by the China Resource Satellite Application Center (http://www.cresda.com/CN). The HS data contain 240 × 240 pixels and 180 bands with 30-m spatial resolution, the spectral range is from 400 to 2500 nm, and the acquisition date is October 7, 2018. The corresponding Sentinel-2A MS data have 720 × 720 × 4 pixels, and the spatial resolution is 10 m and was acquired on October 4, 2018. Besides, the Yellow River Estuary has a complex ecosystem, and the main ground objects include reeds, suaeda, tamarisk, acacia, rivers, ponds, breeding ponds, buildings of different materials, and so on.

4) Poyang Lake Dataset: Poyang Lake HS data are also from the GF-5 HS sensor. The HS data consists of 160 bands with 210 × 210 pixels, and the acquisition date is June 1, 2018. GF-1 MS data from June 25, 2018, are used in the experiment with 840 × 840 pixels. It has four bands with a spatial resolution of 7.5 m. As a typical wetland area, Poyang Lake has complex ground objects and mainly includes rivers, lakes, ponds, wild grasses, floating vegetation, shrubs, reeds, buildings of different materials, and so on.

B. Evaluation Metrics

Since the simulation datasets provide reference images, we perform a quantitative evaluation for the fused images of the simulated datasets according to Wald's protocol [50]–[52]. Specifically, we use SAM [11], the Erreur relative globale adimensionnelle de synthe Ìse (ERGAS) [53], the peak signal-to-noise ratio (PSNR) [54], quality w/ no reference (QNR) [8], and cross correlation (CC) [1]. SAM is used to evaluate the spectral distortion of the fused image, and the ideal value is 0. ERGAS, PSNR, and QNR are used to evaluate the global distortion of the fused image, a larger ERGAS brings about more spectral and spatial distortions, a larger PSNR is the better fusion quality, and the ideal value of QNR is 1. CC is used to evaluate the spatial distortion of the fused images, and the ideal value is 1.

C. Experimental Results

To compare the fusion performance of LOISF, these state-of-the-art benchmark methods are selected, including GSA [55], MTF-GLP [17], CNMF [28], FUSE [56], NLSTF [31], coupled block term decompositions for HS super-resolution with variability (CB-STAR) [57], multiresolution spatial–spectral feature learning (MSSL) [58], Two-CNN [38], and PFCN [40].

1) Simulated Datasets' Results: The fusion results of Chikusei and DFC2018 Houston datasets are shown in Figs. 3 and 4. SAM maps are shown in Fig. 5, which describes the spectral error of these fused results. All methods achieve good visualization results. However, for the Chikusei dataset, MTF-GLP and PFCN achieve severe spectral distortion in the river area. GSA, CNMF, FUSE, NLSTF, CB-STAR,

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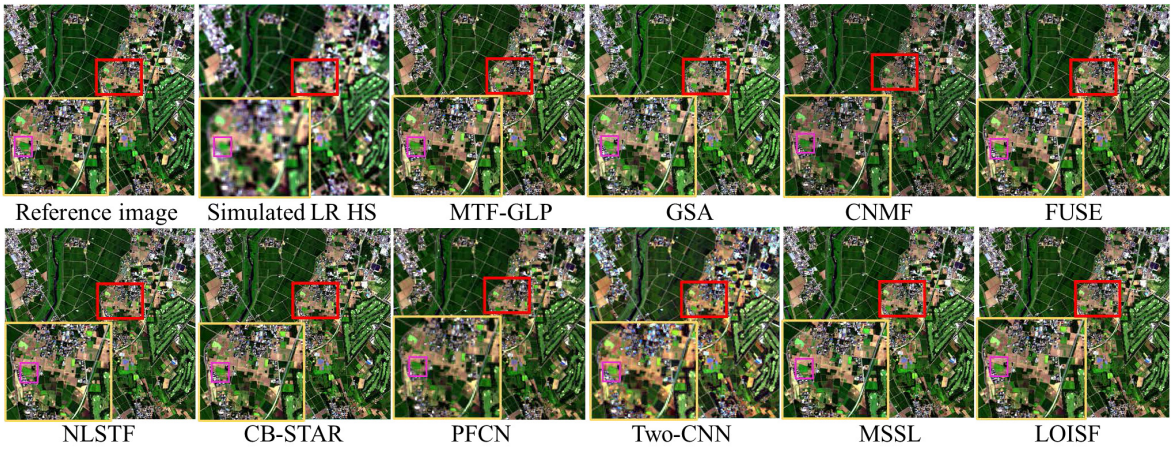

Fig. 3. Fusion results in the Chikusei dataset from different methods.

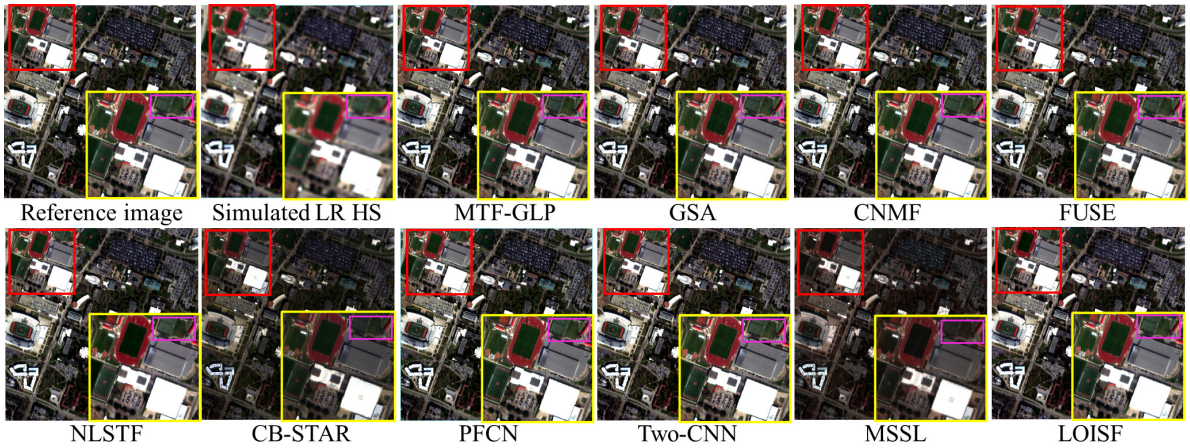

Fig. 4. Fusion results in the DFC2018 Houston dataset from different methods.

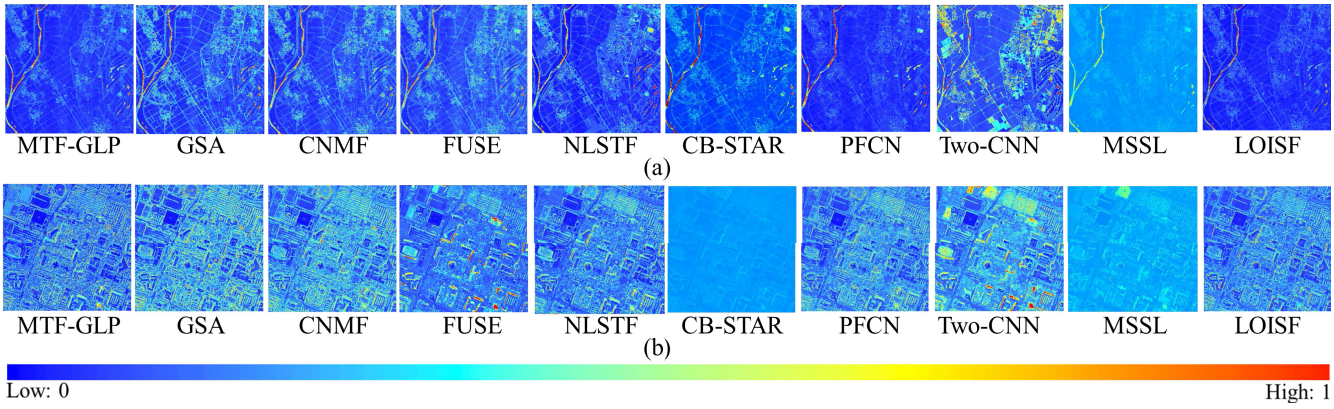

Fig. 5. SAM maps of (a) Chikusei dataset and (b) DFC2018 Houston dataset.

Two-CNN, and MSSL have obvious spectral errors in the building area with different colored roofs. For the DFC2018 Houston dataset, GSA and CNMF have poor spectral fidelity for the black roof and gray roof buildings. FUSE obtains obvious spectral distortion at the edge of the ground objects. MTF-GLP, NLSTF, and PFCN have comparative spectral fidelity. CB-STAR, Two-CNN, and MSSL obtain the worst fusion results in all simulated datasets with global spectral and spatial distortion, and LOISF achieves the best spectral and spatial fidelity by local optimization strategy and strong constraints of band relations. Table II shows the band-averaged values of quantitative evaluation, and Fig. 6

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TABLE II
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE SIMULATED EXPERIMENTS

Methods	SAM		ERGAS		PSNR		CC		QNR	
	Chikusei	DFC2018 Houston	Chikusei	DFC2018 Houston	Chikusei	DFC2018 Houston	Chikusei	DFC2018 Houston	Chikusei	DFC2018 Houston
MTF-GLP	1.5600	1.1070	1.5657	0.8932	41.4526	42.9354	0.9959	0.9981	0.9821	0.9882
GSA	1.5186	1.2062	2.0146	1.1138	39.2944	40.9093	0.9932	0.9972	0.9856	0.9812
CNMF	1.6669	1.0967	1.2494	0.8488	44.0107	46.8704	0.9973	0.9967	0.9768	0.9891
FUSE	2.2925	1.3977	2.9351	1.4312	36.8174	40.3489	0.9899	0.9927	0.9602	0.9784
NLSTF	1.3119	1.0699	1.0797	0.7897	45.2801	46.1782	0.9976	0.9975	0.9899	0.9878
CB-STAR	3.6912	2.0136	4.9621	1.9685	33.5930	38.5636	0.9688	0.9901	0.8958	0.9669
PFCN	1.2933	1.1004	0.9127	0.5426	48.5529	50.0082	0.9972	0.9982	0.9912	0.9955
Two-CNN	7.1281	2.4659	10.8803	1.6758	23.9605	37.4656	0.6896	0.9939	0.7746	0.9422
MSSL	3.7985	2.3465	5.4036	2.5494	32.2822	34.1210	0.9589	0.9807	0.8723	0.9623
LOISF	**1.1228**	**0.8489**	**0.8526**	**0.5346**	**49.2278**	**51.7963**	**0.9984**	**0.9986**	**0.9964**	**0.9981**

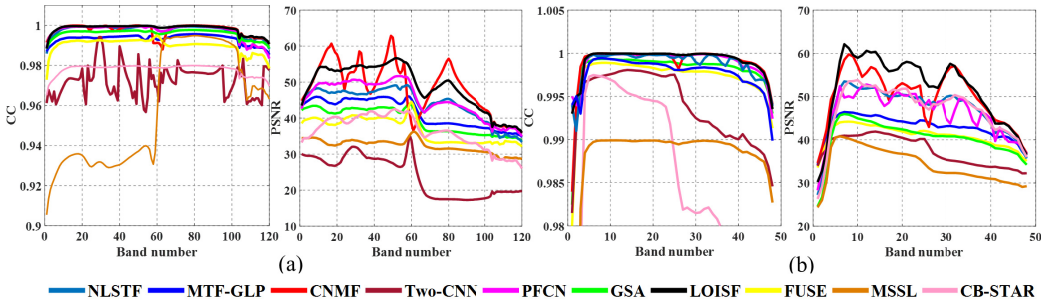

Fig. 6. PSNR and CC of all bands. (a) Chikusei. (b) DFC2018 Houston.

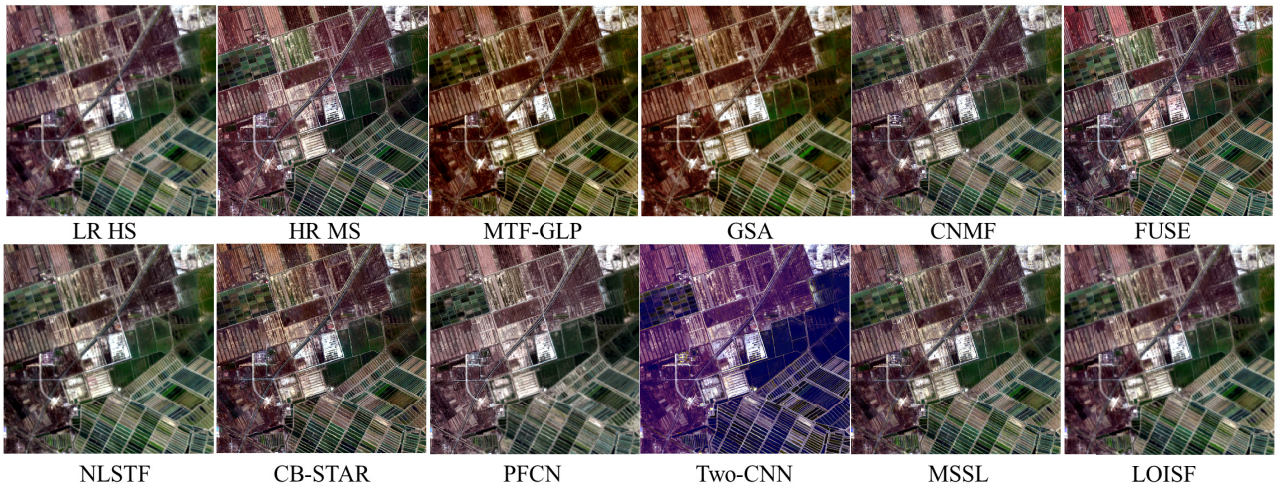

Fig. 7. Fusion results of different methods in the Yellow River Estuary dataset.

shows the quantitative results of each band of the fused images. LOISF obtains the best value among all indicators. PFCN, NLSTF, MTF-GLP, GSA, and CNMF followed by FUSE, CB-STAR, MSSL, and Two-CNN achieve the worst results. Among these methods, GSA causes serious spectral distortion through component replacement and is not suitable for complex underlying surface areas. Machine-learning-based methods could not well extract spectral features of subclass objects and cause serious reconstruction errors. The spectral characteristics have not been changed in MTF-GLP, and the spatial detail injection improves the spatial resolution while retaining smaller spectral distortion. Compared with MSSL and TWO-CNN, PFCN has a small amount of demand for training data and uses Laplacian pyramid decomposition combined with CNN, and sufficient spectral and spatial features greatly promote the fusion accuracy.

2) Real Datasets' Results: The fusion results of Yellow River Estuary are shown in Fig. 7. All methods are obvious for the spatial resolution enhancement of LR HS; however, there are serious spectral differences in these fusion results. CB-STAR, Two-CNN, and MSSL have global spectral distortion. MTF-GLP and GSA have spectral differences with LR HS in

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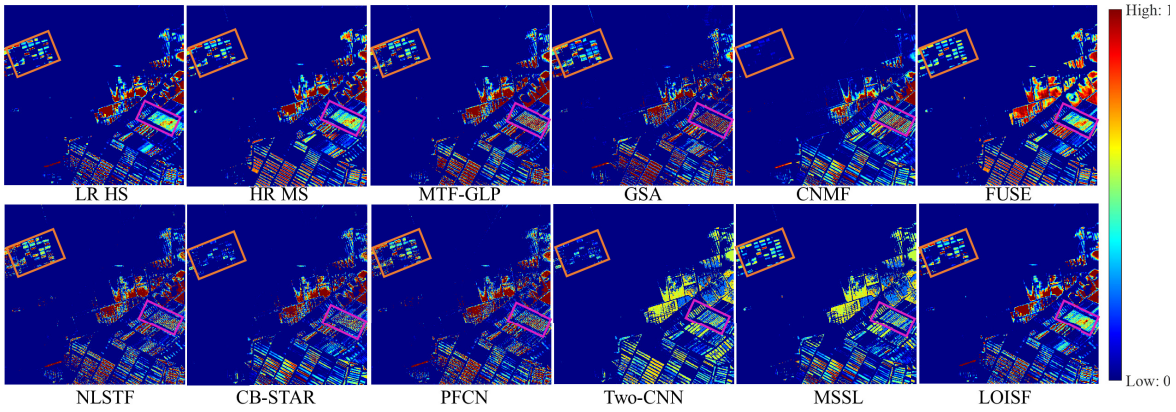

Fig. 8. NDWI of fusion results of different methods in the "Yellow River Estuary" dataset.

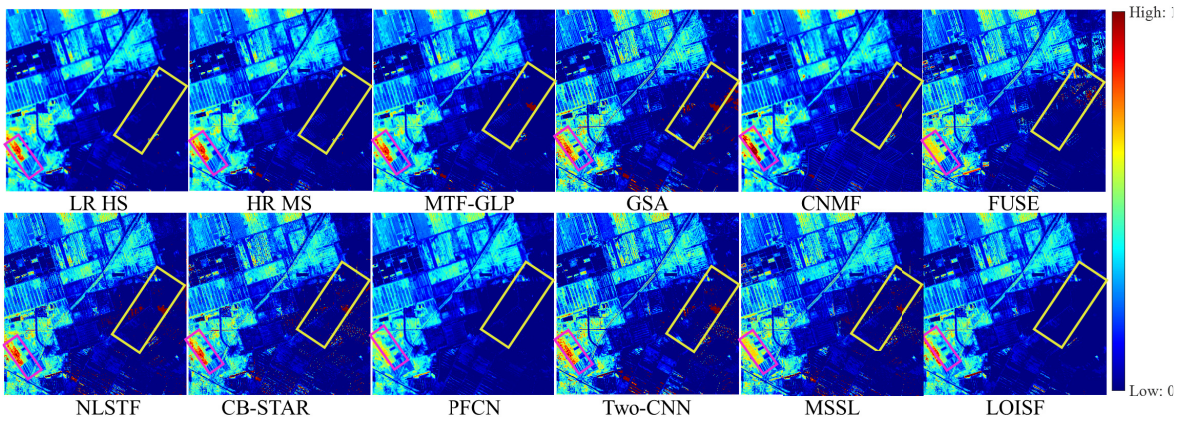

Fig. 9. NDVI of fusion results of different methods in the "Yellow River Estuary" dataset.

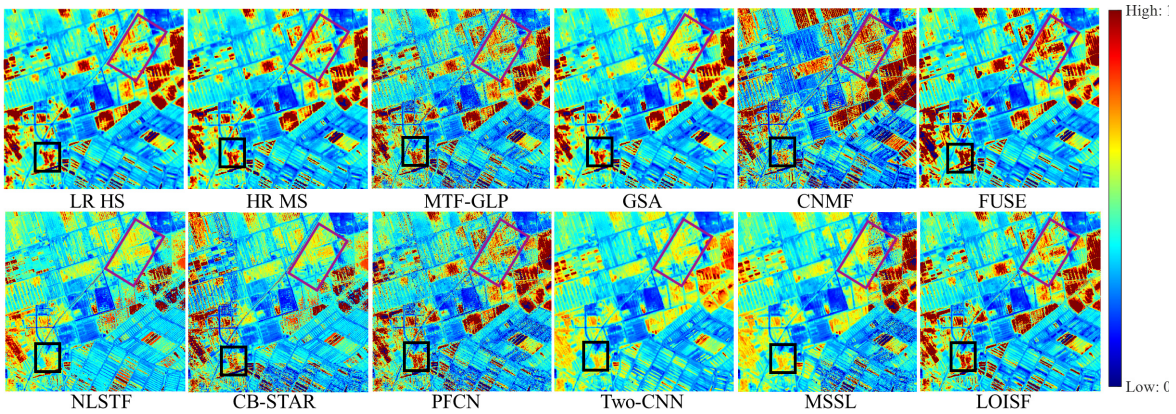

Fig. 10. SNDBI of fusion results of different methods in the "Yellow River Estuary" dataset.

water bodies and farmland areas. FUSE obtains poor spectral fidelity for water bodies and vegetation areas. CNMF, NLSTF, and PFCN achieve better spectral fidelity, and some errors exist in the bare area. LOISF obtains the best spatial and spectral visualization results. These results show that LOISF can obtain comparative fusion performance in different underlying surfaces and is more robust. MTF-GLP, CNMF, and PFCN are suitable for areas with simple objects. MSSL and Two-CNN have a strong demand for training data amount, and other methods have lower robustness.

Figs. 8–10 respectively, show the NDWI, NDVI, and SNDBI indexes of the fused images downsampled by spectrum for "Yellow River Estuary" dataset. LOISF has the most consistent results with HS and MS, followed by PFCN, NLSTF, and MTF-GLP; and GSA, CNMF, CB-STAR, MSSL, Two-CNN, and FUSE have significant differences compared with HS and MS. The fusion results of Poyang Lake datasets are shown in Fig. 11 and are similar to those of the Yellow River Estuary. Figs. 12–14, respectively, show the NDWI, NDVI, and SNDBI indexes of the fused images for "Poyang Lake" dataset.

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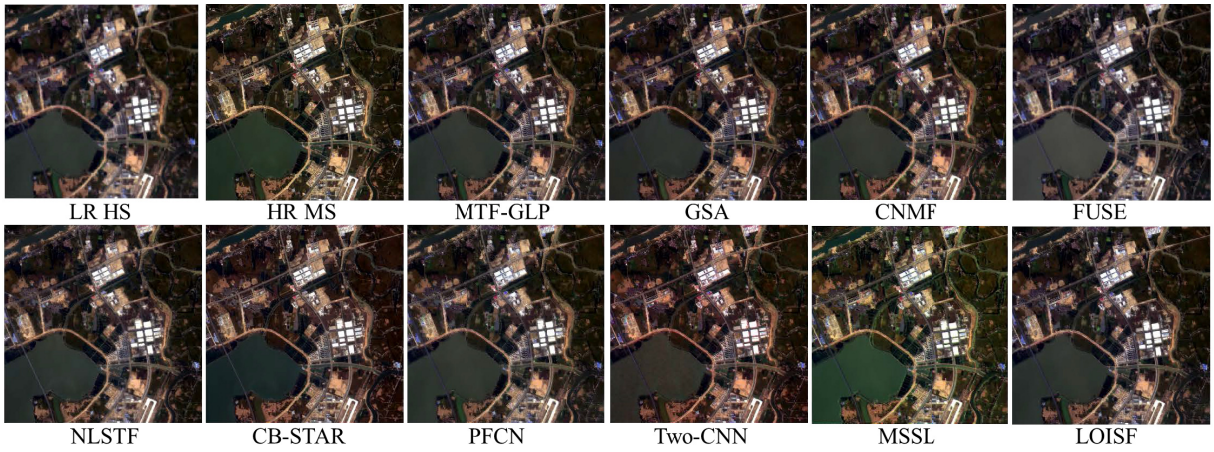

Fig. 11. Fusion results from different methods in the "Poyang Lake" dataset.

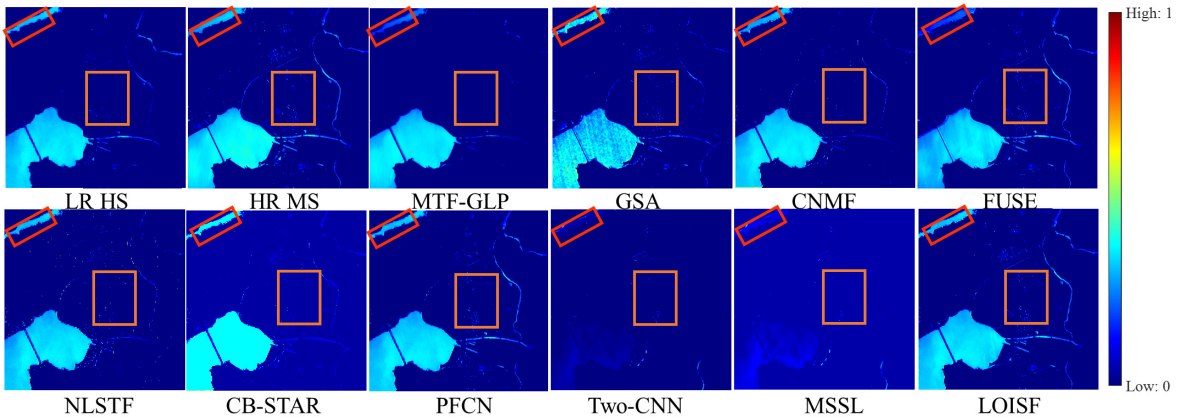

Fig. 12. NDWI of fusion results of different methods in the "Poyang Lake" dataset.

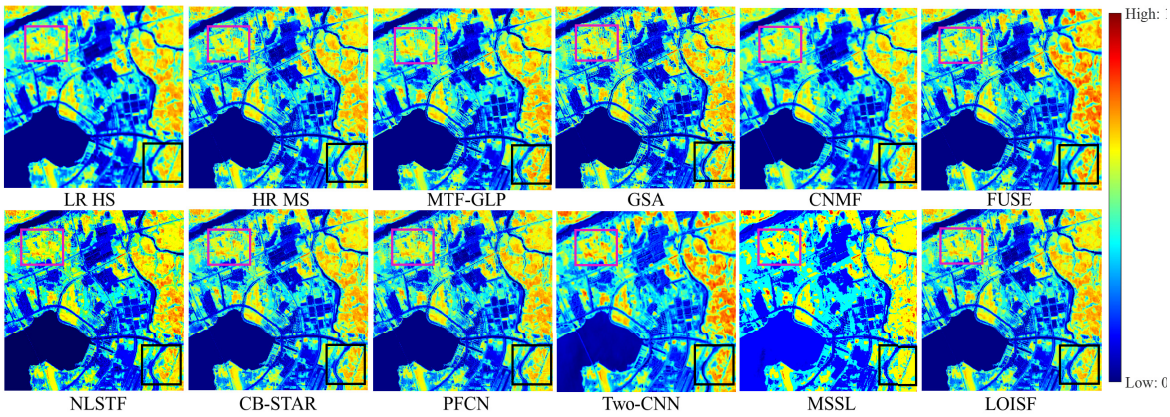

Fig. 13. NDVI of fusion results from different methods in the "Poyang Lake" dataset.

MTF-GLP, GSA, FUSE, MSSL, and Two-CNN have the worst results in NDWI; the errors of FUSE, MSSL, and Two-CNN are the most serious in NDVI; and GSA, CNMF, FUSE, CB-STAR, and Two-CNN have obvious difference compared with LR HS and HR MS. LOISF is still the best. Besides, QNR is used to quantitatively evaluate the fusion results of real datasets. Table III shows that LOISF obtained the best results, followed by MTF-GLP, PFCN, CNMF, and NLSTF, and the fusion quality of other methods is limited.

Fig. 15 plots the spectral curves of removed the continuum of different objects (Tamarix, locust, reed, pond, breeding pond, lotus root pond, red roof building, blue roof building, and bright roof building) in the "Yellow River Estuary" dataset. These curves have obvious differences compared with

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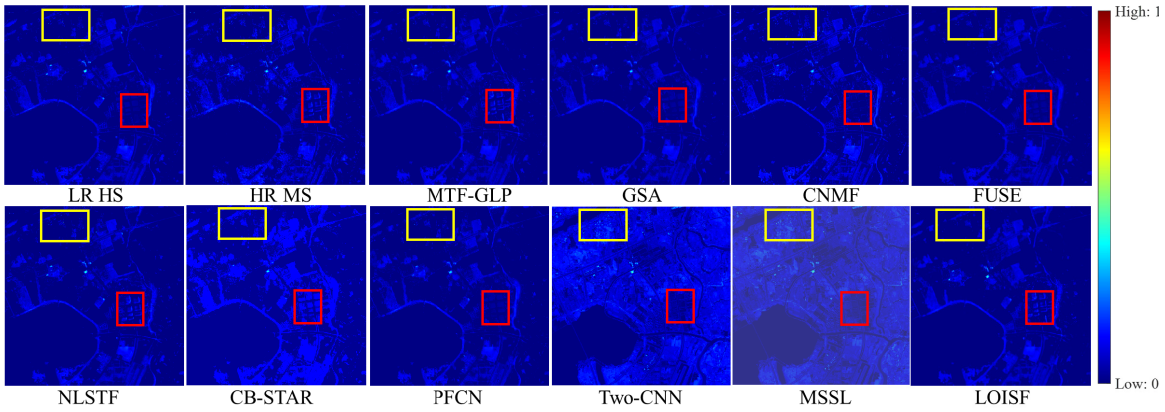

Fig. 14. SNDBI of fusion results from different methods in the "Poyang Lake" dataset.

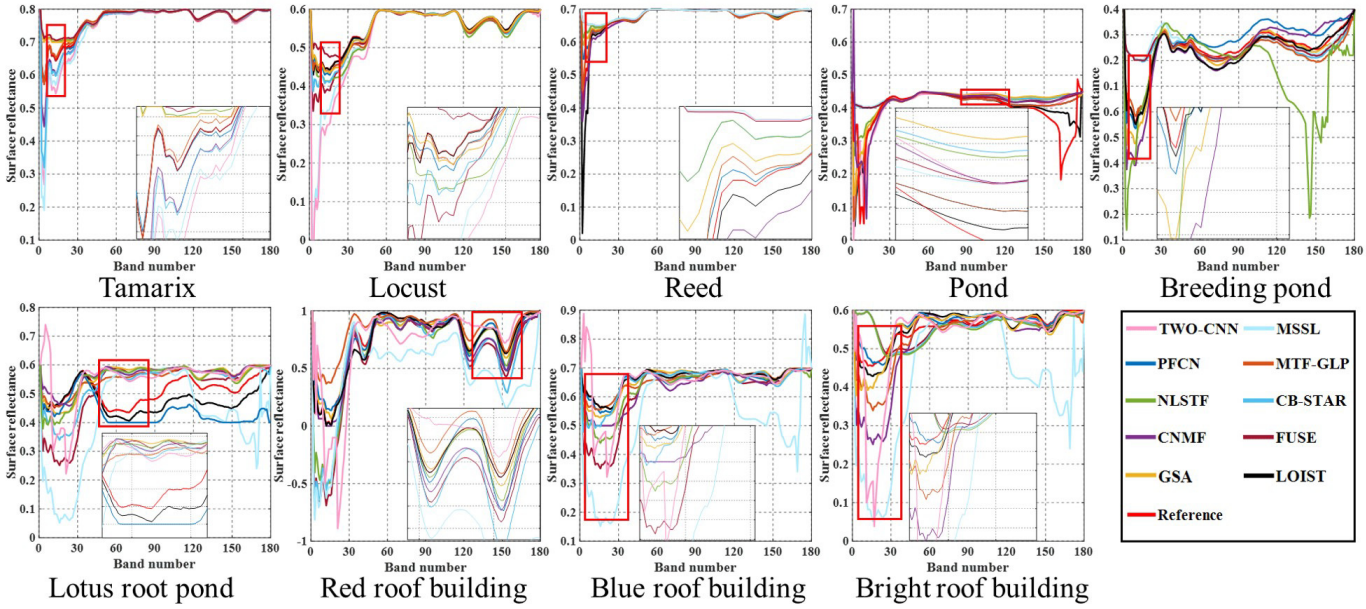

Fig. 15. Spectral differences of different objects in the "Yellow River Estuary" dataset.

TABLE III
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE REAL EXPERIMENTS

Methods	QNR	
	Yellow River Estuary	**Poyang Lake**
MTF-GLP	0.9441	0.9233
GSA	0.9187	0.9201
CNMF	0.9361	0.8992
FUSE	0.9251	0.8824
NLSTF	0.9308	0.9139
CB-STAR	0.8826	0.8660
PFCN	0.9412	0.9116
Two-CNN	0.7169	0.7006
MSSL	0.8678	0.7961
LOISF	**0.9523**	**0.9388**

the reference spectral curves. The spectral curves of LOISF are almost consistent with the reference spectrum curves in all objects. CNMF, MTF-GLP, and GSA are unstable. For example, CNMF maintains better spectral curves for locust, reed, and bright roof buildings. However, the spectral curves of the breeding pond, lotus root pond, red roof building, and blue roof building have serious deviations; the spectral curves of GSA have obvious errors in Tamarix, reed, pond, breeding pond, and red roof building compared with the reference spectrum curves; and MTF-GLP maintains poor spectral curves in Tamarix, reed, corn, and blue roof building. PFCN and NLSTF maintain better spectral curves in all objects compared with other benchmark methods. The worst results are among FUSE, CB-STAR, MSSL, and Two-CNN, and there are serious errors in the spectral curves of all objects.

Fig. 16 plots the spectral curves of removed the continuum from different objects (river, lake, pond, Yegu grass, duckweed, shrub, red roof building, blue roof building, and bright roof building) in the "Poyang Lake" dataset. FUSE, CB-STAR, MSSL, and Two-CNN still have obvious errors in the spectral curves of all objects compared with the reference curves;

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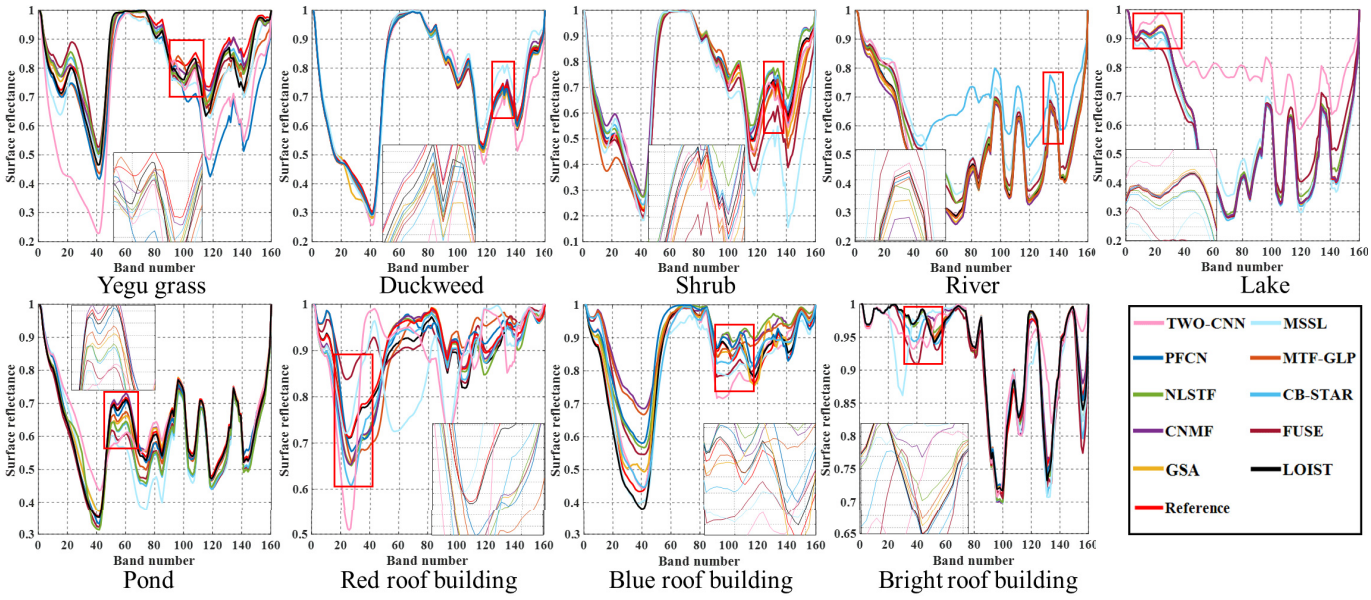

Fig. 16. Spectral differences of different objects in the Poyang Lake dataset.

MTF-GLP, CNMF, NLSTF, and PFCN maintain better spectral curves of the river, lake, Yegu grass, duckweed, shrub, arbor, and bright roof building; and GSA has obvious errors in the spectral curves of the river, pond, duckweed, red roof building, blue roof building, and bright roof building. Although there are some fluctuations, our method still achieves the closest values to the reference curves in all objects.

By comparing spectral curves of ground objects, LOISF better retains the spectral fidelity of subclass objects and effectively reduces spectral errors during the fusion process. Other methods obtain reconstructed data through a global solution; ignoring the spectral distortion caused by local errors, it could not constrain the band relationship of the fused data; and the performance is unstable.

D. Discussion

Matrix factorization-based methods achieve image fusion by reconstructing the spectral matrix of LR HS and spatial matrix of HR MS. However, the current methods obtain HR HS by the global optimal solution. On the one hand, the extracted spectral information is limited when the objects are complicated, and the spectral information of the subclass is underutilized. This unrepresentative spectral information usually causes serious spectral distortion of the reconstructed image. On the other hand, the local optimal solution cannot be obtained, the fusion quality of different objects is quite different, and the relationship between bands of the fused image is not constrained. In this article, we proposed a novel locally optimized HS super-resolution reconstruction method for MS/HS fusion with high fidelity to solve the above problems. The proposed LOISF has fewer parameters and robust performance and outperforms several state-of-the-art methods.

For the LOISF, it is important to set the number of clusters (F). We suggest that: 1) when the coverage area of MS data is larger (>400 km^2) and the spatial resolution of MS data is less than 10 m, $6 \leq F \leq 10$; 2) when the coverage area of MS data is small (≤ 400 km^2) and the spatial resolution of MS data is less than 10 m, $4 \leq F < 6$; 3) when the coverage area of MS data is larger (>400 km^2) and the spatial resolution of MS data is larger than 10 m, $4 \leq F \leq 8$; and 4) when the coverage area of MS data is small (≤ 400 km^2) and the spatial resolution of MS data is larger than 10 m, $4 \leq F \leq 6$. In addition, when there are a large number of fragmented patches in clustered images, mathematical morphology operators should be used for reclustering.

TABLE IV
COMPUTATIONAL TIME OF DIFFERENT METHODS

Methods	Times (S)			
	Chikusei	DFC2018 Houston	Yellow River Estuary	Poyang Lake
MTF-GLP	25.26	20.68	45.34	34.23
GSA	40.82	33.25	67.24	58.45
CNMF	165.65	158.98	201.68	198.87
FUSE	134.25	115.69	159.35	141.23
NLSTF	268.12	249.87	398.88	375.65
Two-CNN	836.46	805.74	1025.98	968.34
PFCN	867.28	816.20	1168.54	997.81
LOISF	216.97	191.63	301.54	258.67

To select an appropriate regularization coefficient φ of different models, the performance of the proposed method has been evaluated. The results on two simulated datasets are shown in Fig. 17, showing that superior indexes are produced and the variations of all evaluation indexes are obvious when φ is from 0.001 to 20, and the changing trends are flat for lager φ in these models. One can see that SAM decreases rapidly, and CC increases rapidly first and then flattens out with the increasing of φ in all models. Considering the overall results, φ_1, φ_2, and φ_3 are set as 15 finally in all datasets.

LOISF was performed on MATLAB 2018a and Intel Core i5 processor with 48-GB RAM. The comparison of computation times for these methods is shown in Table IV. GSA and

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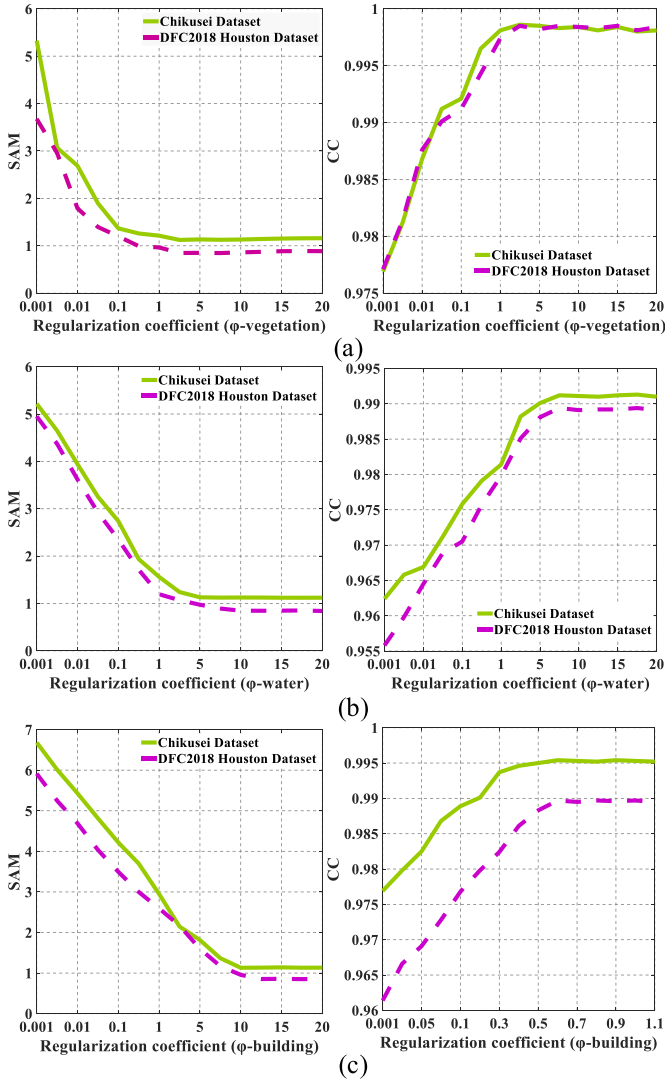

Fig. 17. Performance of the proposed fusion method versus φ. (a) Vegetation fusion model. (b) Water fusion model. (c) Building fusion model.

MTF-GLP have achieved better computing performance, followed by FUSE, CNMF, LOISF, and NLSTF; Two-CNN and PFCN have extremely low computing efficiency. In LOISF, each segmented image needs to be fused, and each fusion involves three loop iterations. Therefore, LOISF has lower computational efficiency. However, LOISF still has comparative advantages compared with deep learning methods and NLSTF.

V. Conclusion

In this article, a novel HS and MS fusion method, i.e., the LOISF, was proposed to obtain a high-fidelity fused image by local space, spectrum, and band relationship constraints. The HESSC method was first used to segment the dataset. Then, we identified the label of each segmented dataset. Finally, we established the fusion model of vegetation area, water area, and building area, respectively, and combined spectral, spatial, and index constraints to reconstruct these segmented images by multiplicative update rules. We verified the proposed method on multiple datasets, and the results show that LOISF achieved better fusion performance compared with the other state-of-the-art methods. In realistic applications, a moderate number of clusters (F) equal to 6 and a moderate regularization coefficient (φ) around 15 are recommended to use to guarantee the good performance of LOISF.

References

[1] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[2] C. M. Gevaert and F. J. García-Haro, "A comparison of STARFM and an unmixing-based algorithm for Landsat and MODIS data fusion," *Remote Sens. Environ.*, vol. 156, pp. 34–44, Jan. 2015.

[3] L. Loncan *et al.*, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[4] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Inf. Fusion*, vol. 46, pp. 102–113, Jun. 2018.

[5] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 3586–3594.

[6] T. N. Carlson and D. A. Ripley, "On the relation between NDVI, fractional vegetation cover, and leaf area index," *Remote Sens. Environ.*, vol. 62, no. 3, pp. 241–252, 1997.

[7] B.-C. Gao, "NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space," *Remote Sens. Environ.*, vol. 58, no. 3, pp. 257–266, Dec. 1996.

[8] G. Vivone *et al.*, "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[9] M. Selva, B. Aiazzi, F. Butera, L. Chiarantini, and S. Baronti, "Hyper-sharpening: A first approach on SIM-GA data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3008–3024, Jun. 2015.

[10] Z. Chen, H. Pu, B. Wang, and G.-M. Jiang, "Fusion of hyperspectral and multispectral images: A novel framework based on generalization of pan-sharpening methods," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 8, pp. 1418–1422, Aug. 2014.

[11] W. Sun, K. Ren, X. Meng, C. Xiao, G. Yang, and J. Peng, "A band divide-and-conquer multispectral and hyperspectral image fusion method," *IEEE Trans. Geosci. Remote Sens.*, early access, Feb. 25, 2021, doi: 10.1109/TGRS.2020.3046321.

[12] W. J. Carper, T. M. Lillesand, and R. W. Kiefer, "The use of intensity-hue-saturation transformations for merging SPOT panchromatic and multispectral image data," *Photogramm. Eng. Remote Sens.*, vol. 56, no. 4, pp. 459–467, Apr. 1990.

[13] A. Y. Kwarteng and P. S. Chavez, Jr., "Extracting spectral contrast in Landsat thematic mapper image data using selective principal component analysis," *Photogramm. Eng. Remote Sens.*, vol. 55, no. 3, pp. 339–348, Mar. 1989.

[14] V. K. Shettigara, "A generalized component substitution technique for spatial enhancement of multispectral images using a higher resolution data set," *Photogram. Eng. Remote Sens.*, vol. 58, no. 5, pp. 561–567, May 1992.

[15] K. Ren, W. Sun, X. Meng, G. Yang, and Q. Du, "Fusing China GF-5 hyperspectral data with GF-1, GF-2 and Sentinel-2A multispectral data: Which methods should be used?" *Remote Sens.*, vol. 12, no. 5, pp. 1–19, 2020.

[16] P. Chavez, S. C. Sides, and J. A. Anderson, "Comparison of three different methods to merge multiresolution and multispectral data Landsat TM and SPOT panchromatic," *Photogramm. Eng. Remote Sens.*, vol. 57, no. 3, pp. 295–303, Mar. 1991.

[17] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Oct. 2002.

[18] G. Vivone, R. Restaino, M. D. Mura, G. Licciardi, and J. Chanussot, "Contrast and error-based fusion schemes for multispectral image pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 930–934, May 2013.

[19] S. Li and B. Yang, "A new pan-sharpening method using a compressed sensing technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 738–746, Feb. 2011.

[20] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Dec. 2000.

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[21] X. Meng, H. Shen, Q. Yuan, H. Li, L. Zhang, and W. Sun, "Pansharpening for cloud-contaminated very high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2840–2854, May 2019.

[22] X. Meng *et al.*, "A large-scale benchmark data set for evaluating pansharpening performance: Overview and implementation," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 18–52, Mar. 2021.

[23] Z. H. Nezhad, A. Karami, R. Heylen, and P. Scheunders, "Fusion of hyperspectral and multispectral images using spectral unmixing and sparse coding," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2377–2389, Jun. 2016.

[24] Q. Wang, Q. Li, and X. Li, "Hyperspectral image superresolution using spectrum and feature context," *IEEE Trans. Ind. Electron.*, vol. 68, no. 11, pp. 11276–11285, Nov. 2021.

[25] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J. Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[26] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C.-W. Chan, and W. Philips, "Spatial-spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[27] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. CVPR*, 2011, pp. 2329–2336.

[28] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[29] W. Wan, W. Guo, H. Huang, and J. Liu, "Nonnegative and nonlocal sparse tensor factorization-based hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8384–8394, Dec. 2020.

[30] K. Zhang, M. Wang, S. Yang, and L. Jiao, "Spatial–spectral-graph-regularized low-rank tensor decomposition for multispectral and hyperspectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1030–1040, Apr. 2018.

[31] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5344–5353.

[32] Y. Xu, Z. Wu, J. Chanussot, P. Comon, and Z. Wei, "Nonlocal coupled tensor cp decomposition for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 348–362, Mar. 2019.

[33] L. Zhang, L. Zhang, D. Tao, and X. Huang, "A modified stochastic neighbor embedding for multi-feature dimension reduction of remote sensing images," *ISPRS J. Photogrammetry Remote Sens.*, vol. 83, pp. 30–39, Sep. 2013.

[34] D. Shen, J. Liu, Z. Wu, J. Yang, and L. Xiao, "ADMM-HFNet: A matrix decomposition-based deep approach for hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, early access, Sep. 24, 2021, doi: 10.1109/TGRS.2021.3112181.

[35] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.

[36] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Ses.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[37] J. Yang, L. Xiao, Y.-Q. Zhao, and J. C.-W. Chan, "variational regularization network with attentive deep prior for hyperspectral-multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, early access, May 27, 272021, doi: 10.1109/TGRS.2021.3080697.

[38] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Hyperspectral and multispectral image fusion via deep two-branches convolutional neural network," *Remote Sens.*, vol. 10, no. 5, p. 800, 2018.

[39] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Multispectral and hyperspectral image fusion using a 3-D-convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 639–643, May 2017.

[40] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Pyramid fully convolutional network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1549–1558, May 2019.

[41] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[42] J. Li *et al.*, "Hyperspectral image super-resolution by band attention through adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4304–4318, Jun. 2020.

[43] B. Zhukov, D. Oertel, F. Lanzl, and G. Reinhackel, "Unmixing-based multisensor multiresolution image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1212–1226, May 1999.

[44] X. X. Zhu and R. Bamler, "A sparse image fusion algorithm with application to pan-sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2827–2836, May 2013.

[45] M. K. Firozjaei *et al.*, "Evaluating the spectral indices efficiency to quantify daytime surface anthropogenic heat island intensity," *Int. Methodol.*, vol. 12, no. 4, pp. 1–12, 2020.

[46] K. R. Shahi, M. Khodadadzadeh, L. Tusa, P. Ghamisi, R. Tolosana-Delgado, and R. Gloaguen, "Hierarchical sparse subspace clustering (HESSC): An automatic approach for hyperspectral image analysis," *Remote Sens.*, vol. 12, no. 15, p. 2421, Jul. 2020.

[47] J. Sevilla, G. Martin, and J. M. P. Nascimento, "Parallel hyperspectral unmixing method via split augmented Lagrangian on GPU," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 626–630, May 2016.

[48] U. Kumar, C. Milesi, R. R. Nemani, S. K. Raja, S. Ganguly, and W. Wang, "Sparse unmixing via variable splitting and augmented Lagrangian for vegetation and urban area classification using Landsat data," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 40, no. 7, p. 59, 2015.

[49] C.-I. Chang, C.-C. Wu, W.-M. Liu, and Y.-C. Ouyang, "A new growing method for simplex-based endmember extraction algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 10, pp. 2804–2819, Oct. 2006.

[50] L. Liu, B. Liu, H. Huang, and A. C. Bovik, "No-reference image quality assessment based on spatial and spectral entropies," *Signal Process., Image Commun.*, vol. 29, no. 8, pp. 856–863, 2014.

[51] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a 'completely blind' image quality analyzer," *IEEE Signal Process. Lett.*, vol. 20, no. 3, pp. 209–212, Nov. 2012.

[52] G. Wang, Z. Wang, K. Gu, L. Li, Z. Xia, and L. Wu, "Blind quality metric of DIBR-synthesized images in the discrete wavelet transform domain," *IEEE Trans. Image Process.*, vol. 29, pp. 1802–1814, 2020.

[53] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio–temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Dec. 2016.

[54] K. Kotwal and S. Chaudhuri, "A Bayesian approach to visualization-oriented hyperspectral image fusion," *Inf. Fusion*, vol. 14, no. 4, pp. 349–360, Oct. 2013.

[55] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS+ pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Sep. 2007.

[56] Q. Wei, N. Dobigeon, and J. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[57] R. A. Borsoi, C. Prevost, K. Usevich, D. Brie, J. C. M. Bermudez, and C. Richard, "Coupled tensor decomposition for hyperspectral and multispectral image fusion with inter-image variability," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 702–717, Apr. 2021.

[58] J. Qu, Y. Shi, W. Xie, Y. Li, X. Wu, and Q. Du, "MSSL: Hyperspectral and panchromatic images fusion via multiresolution spatial–spectral feature learning networks," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 25, 2021, doi: 10.1109/TGRS.2021.3066374.

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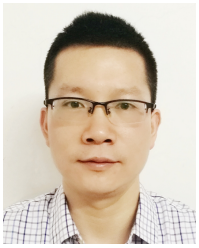

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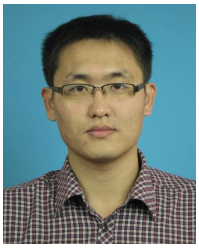

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